What is DOM?

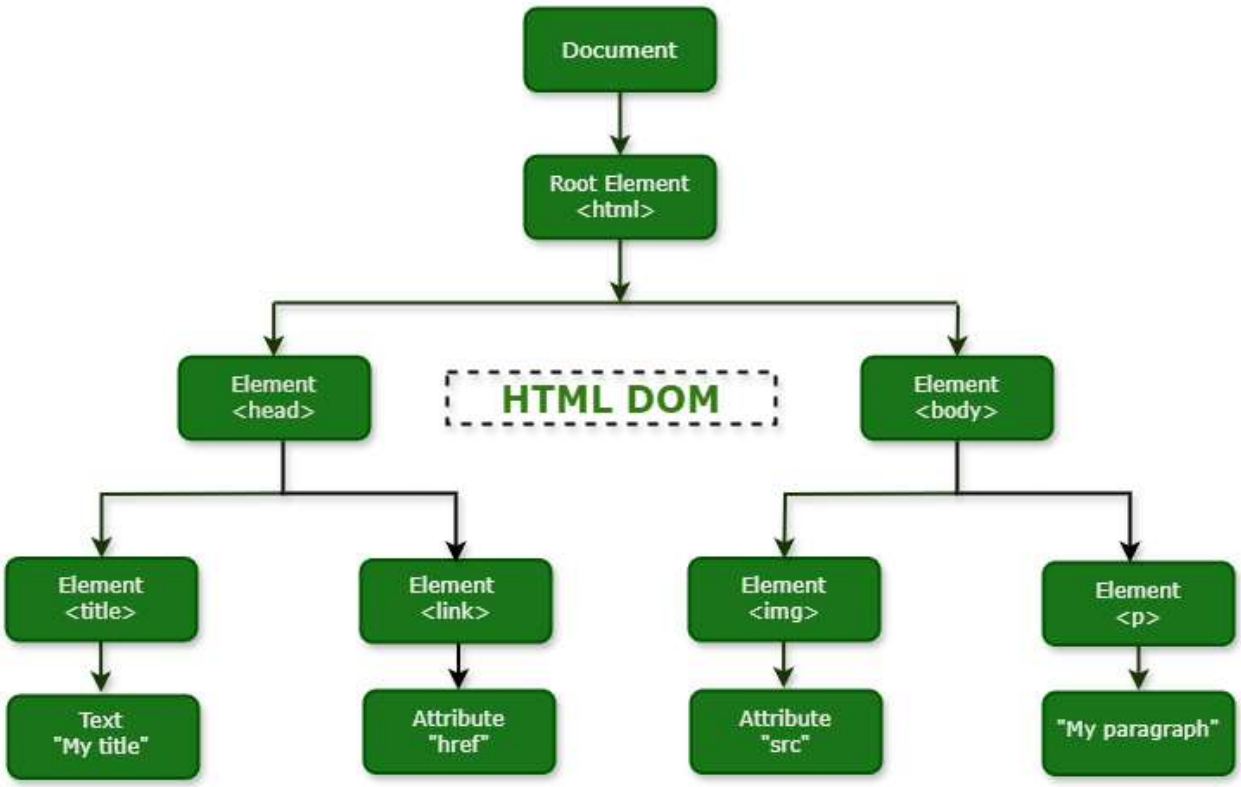
📌 DOM = Document Object Model

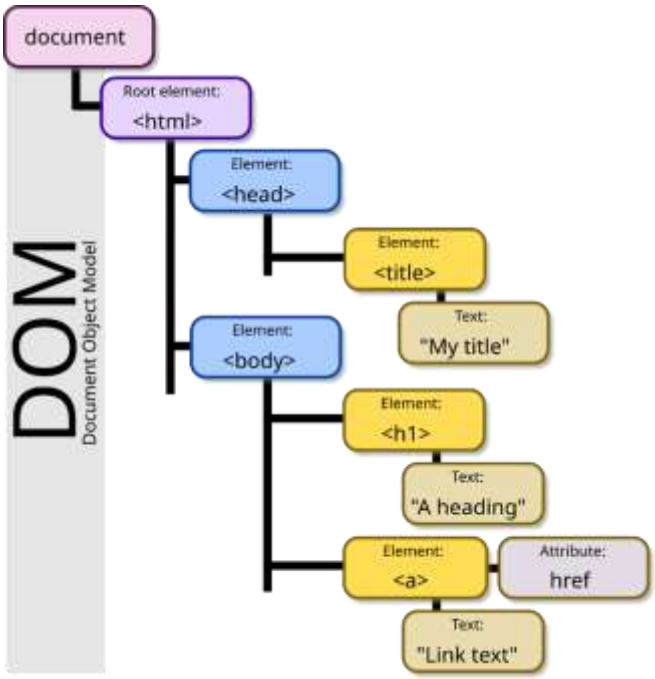
The **DOM** is a **programming interface** for HTML

It represents the webpage as a **tree structure of objects**, where:

- Each **HTML tag** → becomes a **Node**
- Each **attribute** → becomes a **property**
- Each **text** → becomes a **text node**

👉 The browser automatically creates the DOM when it loads an HTML page.





Example HTML

```html
HTML

<!DOCTYPE html>
<html>
  <head>
    <title>My Page</title>
  </head>
  <body>
    <h1>Hello</h1>
    <p>Welcome to DOM</p>
  </body>
</html>
```

🧠 How Browser Sees It (DOM Tree)

```
Code

Document
└── html
    ├── head
    │   └── title
    │       └── "My Page"
    └── body
        ├── h1
        │   └── "Hello"
        └── p
            └── "Welcome to DOM"
```

Every element becomes an object that JavaScript can manipulate.

What is JavaScript DOM?

📌 **JavaScript DOM = Using JavaScript to Access & Manipulate the DOM**

JavaScript interacts with the DOM through the **document object**.

console.log(document);

This gives access to the entire webpage structure.

Why DOM is Important?

Without DOM:

- HTML is static
- No interaction

With DOM:

- Change content dynamically
- Change styles dynamically
- Add/remove elements
- Handle events (click, input, submit)
- Build interactive web applications

How DOM Works with HTML (Step-by-Step Flow)

Browser Process:

1. Browser loads HTML
2. Converts HTML → DOM Tree
3. JavaScript accesses DOM
4. JavaScript modifies DOM
5. Browser re-renders updated content

DOM Selection Methods

Method	**Description**
document.getElementById()	Select by ID
document.getElementsByClassName()	Select by class
document.getElementsByTagName()	Select by tag
document.querySelector()	Select first matching CSS selector
document.querySelectorAll()	Select all matching CSS selector

DOM Properties vs Methods

Properties

- innerHTML
- textContent
- style
- value
- className

Methods

- getElementById()

- querySelector()
- createElement()
- appendChild()
- remove()

DOM Traversal

📌 **What is DOM Traversal?**

DOM Traversal means **navigating between nodes in the DOM tree** — moving from:

- Parent → Child
- Child → Parent
- Sibling → Sibling

🧠 Related Traversal Properties

Property	**Meaning**
parentNode	Immediate parent
parentElement	Same as parentNode (for elements)
children	Only element children
childNodes	All node types
firstElementChild	First element child
lastElementChild	Last element child
nextElementSibling	Next element
previousElementSibling	Previous element